前期 **理系**

2025年度入学試験学力検査問題

英　　語

〔経済経営学部：経済経営学科 数理区分，理学部：化学科，生命科学科，
都市環境学部(都市政策科学科 文系区分を除く)，　　60分
システムデザイン学部(電気電子工学科を除く)〕

答案用紙　　**1枚**

注　　意

1. 監督員の合図があるまで，問題の内容を見てはいけません。
2. 答案用紙に，正しく記入・マークされていない場合は，採点できないことがあります。
3. 解答には黒鉛筆またはシャープペンシルを使用し，答案用紙の解答欄にマークしなさい。例えば，［10］と表示のある問いに対して③と解答する場合は，次の(例)のように**解答番号10の解答欄の③にマーク**しなさい。

(例)

解答番号	解答欄				
	1	2	3	4	5
10	①	②	●	④	⑤

4. 試験中に不鮮明な印刷等に気付いた時は，手をあげて監督員に申し出てください。
5. 答案用紙を切り取ったり，持ち帰ったりしてはいけません。
6. 問題冊子の余白は利用可能ですが，どのページも切り離してはいけません。
7. 問題冊子は，持ち帰ってください。また，試験終了時刻まで退室できません。

第１問 次の英文を読み，設問に答えなさい。なお，設問の都合で本文の段落に 1 ～ 13 の番号を付してあります。

1 A new study is the first-ever to identify the genes for creativity in Homo sapiens that distinguish modern humans from chimpanzees and Neanderthals*1. The research identified 267 genes that are found only in modern humans and likely play an important role in the (a) of the behavioral characteristics that set apart Homo sapiens, including creativity, self-awareness, cooperativeness, and healthy longevity. The study, led by an international and interdisciplinary team of researchers from the American Museum of Natural History and Washington University among other institutions, is published today in the journal *Molecular Psychiatry*.

2 "One of the most fundamental questions about human nature is what sparked the explosive emergence of creativity in modern humans in the period just before and after their widespread dispersal from Africa and the related extinction of Neanderthals and other human relatives," said study co-author Ian Tattersall, curator emeritus in the American Museum of Natural History's Division of Anthropology. "Major controversies persist about the basis for human creativity in art and science, as well as about potential differences in cognition, language, and personality that distinguish modern humans from extinct hominids. This new study is the result of a truly pathbreaking use of genomic methodologies to enlighten us about the mechanisms underpinning our (b)."

3 Modern humans demonstrate remarkable creativity compared to their closest living relatives, the great apes (chimpanzees, gorillas, orangutans, and their immediate ancestors), including innovativeness, flexibility, depth of planning, and related cognitive abilities for symbolism and self-awareness that also enable spontaneous generation of narrative art and language. But the genetic basis for the emergence of creativity in modern humans remains a

mystery, even after the recovery of full-genome data for both chimpanzees and our extinct close relatives the Neanderthals.

[4] "It has been difficult to identify the genes that led to the emergence of human creativity before now because of the large number of changes in the human genome after it diverged from the common ancestor of humans and chimpanzees around 10 million years ago, as well as uncertainty about the functions of those changes," said Robert Cloninger, a psychiatrist and geneticist at Washington University in St. Louis, and the lead author of the study. "(c), we began our research by first identifying the way the genes that influence modern human personality are organized into coordinated systems of learning that have allowed us to adapt flexibly and creatively to changing life conditions."

[5] The team led by Cloninger had previously identified 972 genes that regulate gene expression for human personality, which is comprised of three nearly separate networks for learning and memory. One, for regulating emotional reactivity — emotional drives, habit learning, social attachment, conflict resolution — emerged in monkeys and apes about 40 million years ago. The second, which regulates intentional self-control — self-directedness and cooperation for mutual benefit — emerged a little less than 2 million years ago. A third one, for creative self-awareness, emerged about 100,000 years ago.

[6] In the latest study, the researchers discovered that 267 genes from this larger group are found only in modern humans and not in chimpanzees or Neanderthals. These uniquely human genes code for the self-awareness brain network and also regulate processes that allow Homo sapiens to be creative in narrative art and science, to be more prosocial*2, and to live longer lives through greater resistance to aging, injury, and illness than the now-extinct hominids they (d) replaced.

[7] Genes regulating emotional reactivity were nearly the same in humans, Neanderthals, and chimps. And Neanderthals were about midway between

chimps and Homo sapiens in their genes for self-control and self-awareness.

[8] "We found that the adaptability and well-being of Neanderthals was about 60 to 70 percent of that of Homo sapiens, which means that (e) the difference in fitness between them was large," Cloninger said. "After the more creative, sociable, and physically resilient Homo sapiens migrated out of Africa between 65,000 and 55,000 years ago, they displaced Neanderthals and other hominids, who all became extinct soon after 40,000 years ago."

[9] The genes that distinguish modern humans from Neanderthals and chimpanzees are nearly all regulatory genes made of RNA, not protein-coding genes made of DNA.

[10] "The protein-coding genes of Homo sapiens, Neanderthals, and chimps are nearly all the same, and what distinguishes these species is the regulation of the expression of their protein-coding genes by the genes found only in humans," said co-author Igor Zwir, a computer scientist at Washington University School of Medicine and the University of Granada. "We found that the regulatory genes unique to modern humans were constituents of clusters together with particular protein-coding genes that are overexpressed in the human brain network for self-awareness. The self-awareness network is essential to the physical, mental, and social well-being of humans because it provides the insight to regulate our habits in accord with our goals and values."

[11] The researchers determined that the genes unique to modern humans were selected because of advantages tied to greater creativity, prosocial behavior, and healthy longevity. Living longer, healthier lives and being more prosocial and (f) altruistic allowed Homo sapiens to support their children, grandchildren, and others in their communities throughout their lives in diverse and sometimes harsh conditions. And being more innovative than other hominids allowed humans to adapt more flexibly to unpredictable climatic fluctuations.

[12] "In the bigger picture, this study helps us understand how we can effectively respond to the challenges that modern humans currently face,"

Tattersall said. "Our behavior is not fixed or determined by our genes. Indeed, human creativity, prosociality, and healthy longevity emerged in the context of the need to adjust rapidly to harsh and diverse conditions and to communicate in large social groups."

13 Added co-author Coral del Val of the University of Granada, "Now, we face similar challenges to which we must also respond creatively, as we did originally. Unfortunately, when we are exposed to conditions of fear, conflict, inequity, abuse or neglect, our self-awareness is impaired, which diminishes our ability to use our potential for creativity and to achieve well-being. Learning more about the regulatory genes unique to modern humans may help us to promote human well-being as we face these new environmental and social challenges."

出典：American Museum of Natural History,
"Creativity and community: How modern humans overcame the Neanderthals,"
ScienceDaily (April 21, 2021)
(一部改変)

*1 Neanderthals：ネアンデルタール人

*2 prosocial：社会性のある

問1 空欄(a)に入れるのにもっとも適切なものを，次の①～⑤のうちから一つ選びなさい。 1

① evolution

② transformation

③ expression

④ diversity

⑤ stability

問 2　空欄(b)に入れるのにもっとも適切なものを，次の①～⑤のうちから一つ選びなさい。 [2]

① appearance

② resemblance

③ uniqueness

④ loneliness

⑤ consciousness

問 3　空欄(c)に入れるのにもっとも適切なものを，次の①～⑤のうちから一つ選びなさい。 [3]

① Otherwise

② Instead

③ Besides

④ Therefore

⑤ Meanwhile

問 4　下線部(d)の指すものを，次の①～⑤のうちから一つ選びなさい。 [4]

① genes unique to humans

② Homo sapiens

③ Neanderthals

④ narrative art and science

⑤ aging, injury, and illness

問5　下線部(e)の文意にもっとも近いものを，次の①～⑤のうちから一つ選びなさい。 [5]

① Homo sapiens were larger in size than Neanderthals.

② Homo sapiens were more corruptible than Neanderthals.

③ Homo sapiens were in poorer surroundings than Neanderthals.

④ Homo sapiens were more adaptive than Neanderthals.

⑤ Homo sapiens were weaker than Neanderthals.

問6　[9]～[10]段落を読んで，現生人類に固有の遺伝子についての記述のうち正しいものを，次の①～⑤のうちから一つ選びなさい。 [6]

① The protein-coding genes made of DNA regulate the genes made of RNA.

② The protein-coding genes made of RNA regulate the genes made of DNA.

③ The genes made of DNA regulate the protein-coding genes made of RNA.

④ The genes made of RNA regulate the protein-coding genes made of DNA.

⑤ The genes made of RNA regulate the protein-coding genes made of RNA.

問7　下線部(f)とほぼ同じ意味となるものを，次の①～⑤のうちから一つ選びなさい。 [7]

① calm

② cautious

③ reliable

④ innocent

⑤ selfless

問8　本文の内容にあてはまらないものを，次の①～⑤のうちから一つ選びなさい。 8

① We flexibly adapted to climatic changes.

② We acquired a longer lifespan than other hominids.

③ We left Africa due to the extinction of Neanderthals.

④ We successfully communicated in large groups.

⑤ We helped each other to raise our offspring.

問9　本文の内容と一致するものを，次の①～⑤のうちから一つ選びなさい。 9

① Humans, Neanderthals, and chimps had nearly the same ability to cooperate.

② Neanderthals were replaced by humans because they lacked the ability to control their emotions.

③ Humans and other hominids must have coexisted if the latter had a more flexible mind.

④ Before the genome analysis, Neanderthals were thought to have been as creative as humans.

⑤ Humans started to take a unique path of intellectual development about 100,000 years ago.

問10　本文のタイトルとしてもっとも適切なものを，次の①～⑤のうちから一つ選びなさい。 10

① The genes only found in humans contributed to upright walking.

② The genes unique to Homo sapiens regulate their way of life.

③ The newly identified genes made humans distinct from closely related species.

④ The genes for self-control changed the origin of modern human beings.

⑤ The creative genes were essential for humans to build civilization.

英語の試験問題は次ページに続く。

第2問 次の英文を読み，設問に答えなさい。

The 19th century landscape paintings hanging in London's Tate Britain museum looked awfully familiar to climate physicist Anna Lea Albright. Artist Joseph Mallord William Turner's signature way of shrouding his vistas in fog and smoke reminded Albright of her own research tracking air pollution.

"I started wondering if there was a (a)," says Albright, who had been visiting the museum on a day off from the Laboratory for Dynamical Meteorology in Paris. After all, Turner — a forerunner of the impressionist*1 movement — was painting as Britain's industrial revolution gathered steam, and a growing number of belching*2 manufacturing plants earned London the nickname "The Big Smoke."

Turner's early works, such as his 1814 painting *Apullia in Search of Appullus*, were rendered in sharp details. Later works, like his celebrated 1844 painting *Rain, Steam and Speed — the Great Western Railway*, embraced a dreamier, fuzzier aesthetic.

Perhaps, Albright thought, this burgeoning painting style wasn't a purely artistic phenomenon. Perhaps Turner and his successors painted (b): their environs becoming more and more obscured by smokestack haze.

To find out how much realism there is in impressionism, Albright teamed up with Harvard University climatologist Peter Huybers, who's an expert in reconstructing pollution before instruments existed to closely track air quality. Their analysis of nearly 130 paintings by Turner, Paris-based impressionist Claude Monet and several others tells a tale of two modernizing cities.

Low contrast and whiter hues*3 are hallmarks (c) of the impressionist style. They are also hallmarks of air pollution, which can affect how a distant scene looks to the naked eye. Tiny airborne particles, or aerosols, can absorb or scatter light. That makes the bright parts of objects appear dimmer while also shifting the entire scene's color toward neutral white.

The artworks that Albright and Huybers investigated, which span from the late 1700s to the early 1900s, decrease in contrast as the 19th century progresses. (d) That trend tracks with an increase in air pollution, estimated from historical records of coal sales, Albright and Huybers report in the February 7 *Proceedings of the National Academy of Sciences.*

"Our results indicate that [19th century] paintings capture changes in the optical environment associated with increasingly polluted atmospheres during the industrial revolution," the researchers write.

Albright and Huybers distinguished art from aerosol by first using a mathematical model to analyze the contrast and color of 60 paintings that Turner made between 1796 and 1850 as well as 38 Monet works from 1864 to 1901. They then compared the findings to sulfur dioxide*4 emissions over the century, estimated from the trend in the annual amount of coal sold and burned in London and Paris. When sulfur dioxide reacts with molecules in the atmosphere, aerosols form.

As sulfur dioxide emissions increased over time, the amount of contrast in both Turner's and Monet's paintings decreased. However, paintings of Paris that Monet made from 1864 to 1872 have much higher contrast than Turner's last paintings of London made two decades earlier.

The difference, Albright and Huybers say, can be attributed to the (e) start of the industrial revolution in France. Paris' air pollution level around 1870 was about what London's was when Turner started painting in the early 1800s. It confirms that the similar progression in their painting styles can't be (f) chalked up to coincidence but that it is guided by air pollution, the pair conclude.

The researchers also analyzed the paintings' visibility, or the distance at which an object can be clearly seen. Before 1830, the visibility in Turner's paintings averaged about 25 kilometers, the team found. Paintings made after 1830 had an average visibility of about 10 kilometers. Paintings made by Monet in London around 1900, such as *Charing Cross Bridge*, have a visibility of less than five

kilometers. That's (g) to estimates for modern-day megacities such as Delhi and Beijing, Albright and Huybers say.

To strengthen their argument, the researchers also analyzed 18 paintings from four other London- and Paris-based impressionists. Again, as outdoor air pollution increased over time, the contrast and visibility in the paintings decreased, the team found. What's more, the decrease seen in French paintings (h) lagged behind the decrease seen in British ones.

Overall, air pollution can explain about 61 percent of contrast differences between the paintings, the researchers calculate. In that respect, "different painters will paint in a similar way when the environment is similar," Albright says. "But (i) I don't want to overstep and say: Oh, we can explain all of impressionism."

出典：Bas den Hond,

"Air pollution made an impression on Monet and other 19th century painters,"

Science News (February 26, 2023)

(一部改変)

*1 impressionist：印象派の　*2 belch：(煙が)噴出する

*3 hue：色のぼかし　*4 sulfur dioxide：二酸化硫黄

問1　空欄(a)に入れるのにもっとも適切なものを，次の①～⑤のうちから一つ選びなさい。 11

① solution

② connection

③ question

④ hint

⑤ secret

問 2　空欄(b)に入れるのにもっとも適切なものを，次の①～⑤のうちから一つ選びなさい。 12

① just what they wanted

② precisely what they imagined

③ exactly what they saw

④ even what they caused

⑤ definitely what they created

問 3　下線部(c)とほぼ同じ意味となるものを，次の①～⑤のうちから一つ選びなさい。 13

① features

② downsides

③ benefits

④ remains

⑤ legacies

問 4　下線部(d)の内容にもっとも近いものを，次の①～⑤のうちから一つ選びなさい。 14

① The paintings were becoming colorful in the 19th century.

② The paintings were becoming more blurry as the 19th century went on.

③ The paintings were becoming more naturalistic in the 19th century.

④ The tones of the paintings were becoming clearer as the 19th century went on.

⑤ The styles of the paintings were becoming innovative in the 19th century.

問 5　空欄(e)に入れるのにもっとも適切なものを，次の①～⑤のうちから一つ選びなさい。 15

① much faster

② even earlier

③ no later

④ little sooner

⑤ much slower

問 6　下線部(f)とほぼ同じ意味となるものを，次の①～⑤のうちから一つ選びなさい。 16

① a change for

② a condition of

③ the result of

④ the key to

⑤ the theory of

問 7　空欄(g)に入れるのにもっとも適切なものを，次の①～⑤のうちから一つ選びなさい。 17

① obvious

② outstanding

③ particular

④ similar

⑤ indifferent

問 8　下線部(h)とほぼ同じ意味となるものを，次の①～⑤のうちから一つ選びなさい。 18

① delayed

② was followed by

③ brought about

④ came after

⑤ was caught up with

問 9　下線部(i)の文意としてもっとも適切なものを，次の①～⑤のうちから一つ選びなさい。 19

① I don't assert that impressionist paintings are just a record of air pollution.

② I don't support the idea that air pollution influenced impressionism in some way.

③ I don't insist that there should be more scientific analysis in the study of impressionism.

④ I don't recommend that researchers of impressionist paintings should make records of air pollution.

⑤ I don't adopt the view that the scenes impressionists painted were largely imaginary.

問10　本文の内容と一致するものを，次の①～⑤のうちから一つ選びなさい。

20

① Impressionist painters seem to have tackled environmental problems in those days.

② Impressionist painters were ahead of the times both in artistic styles and perspectives.

③ Impressionist paintings showed a gradual development of a new art movement.

④ Impressionism can be considered a landmark in the history of industrial revolution.

⑤ Impressionism may have derived from the environmental conditions in which the painters lived.